# AGE 316
# MICROCOMPUTER SYSTEMS

LECTURE 5: VARIABLES, DATA TYPES & CONTROL FLOW

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# Recap: Lecture 4

- **Sketch Structure:** The two required functions.
- **setup():** Runs once (Initialization).
- **loop():** Runs forever (Main Logic).
- **Syntax:** Semicolons are mandatory!

```
voi  setu () { ... }
d    p
voi  loo () { ... }
d    p
```

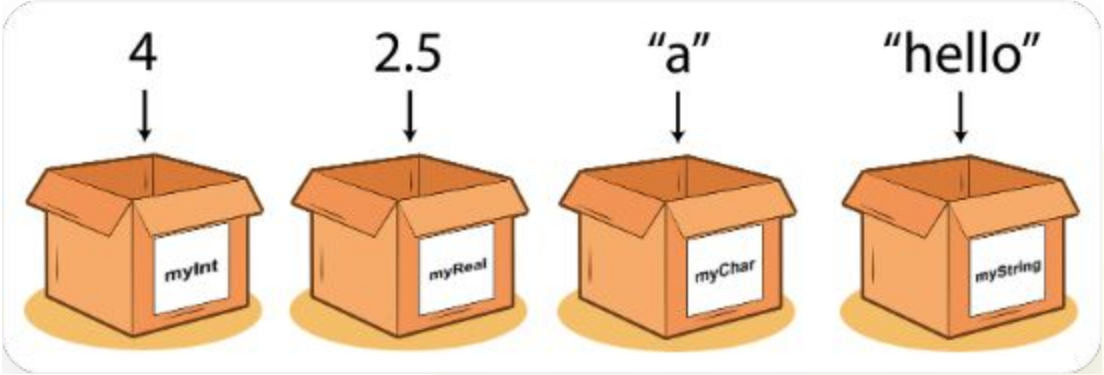
# What is a Variable?

## The "Box" Analogy

Imagine a variable as a storage box.

- **Name:** The label on the box (e.g., myScore).
- **Value:** What is inside the box (e.g., 10).
- **Type:** The shape/size of the box (e.g., can it hold integers or text?).

> **Key Concept:** You can change the *contents* (value), but the *label* (name) stays the same.



# Creating Variables

In C++, you must **Declare** a variable before you use it.

```
// Syntax: Type Name =
Value;
int ledPi  = 1 ;
int delayTime = 1000;
```



**1. Type**

What kind of data? (int)



**2. Name**

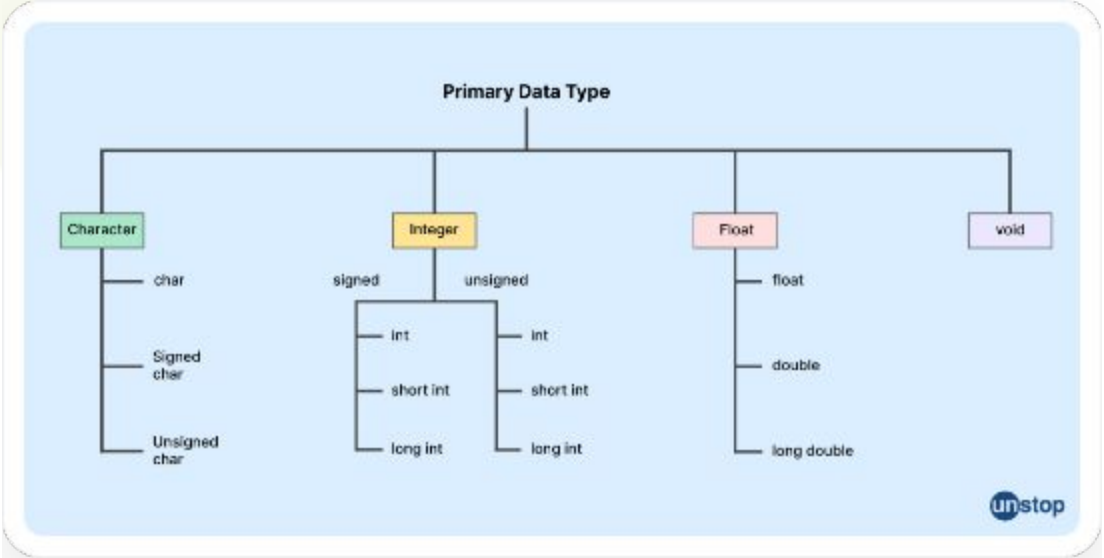
The identifier. (ledPin)



**3. Value**

The initial data. (13)

# Data Types: The "Shapes" of Data




Computer memory is finite. Different data types use different amounts of space.

`int` `float` `bool` `char` `String`

`long` `double`

# Numbers: Int vs. Float

## int (Integer)

Whole numbers only.

```
int count = 5;
int temp = -10;
```

Range: -32,768 to 32,767 (on Uno)

## float (Floating Point)

Numbers with decimals.

```
float pi = 3.14;
float temp = 26.5;
```

Range: Huge, but less precise.

# Logic & Characters

**boolean (bool)**

Holds only two values: true or false (1 or 0).

*Example:* bool isRunning = true;

**char**

A single character in single quotes.

*Example:* char grade = 'A';

**String**

Text string in double quotes (Technically an object, not a primitive).

*Example:* String msg = "Hello World";

# Variable Scope: Global vs. Local

```
int globalVar = 10; // Visible Everywhere

void setup() {
  int localVar = 5; // Only inside setup
}

void loop() {
  // Can use globalVar
  // CANNOT use localVar
}
```

## Where does it live?

**Global:** Declared outside functions. Accessible by *setup* AND *loop*.

**Local:** Declared inside a function (or block). Destroyed when the function ends.

> **Tip:** Use Global for pin numbers. Use Local for temporary calculations.

# Doing Math (Arithmetic)

| + | - | * | / |
| --- | --- | --- | --- |
| Addition | Subtraction | Multiplication | Division |
| sum = a + b; | diff = a - b; | prod = a * b; | div = a / b; |

**Modulo Operator (%):** Returns the remainder of division.

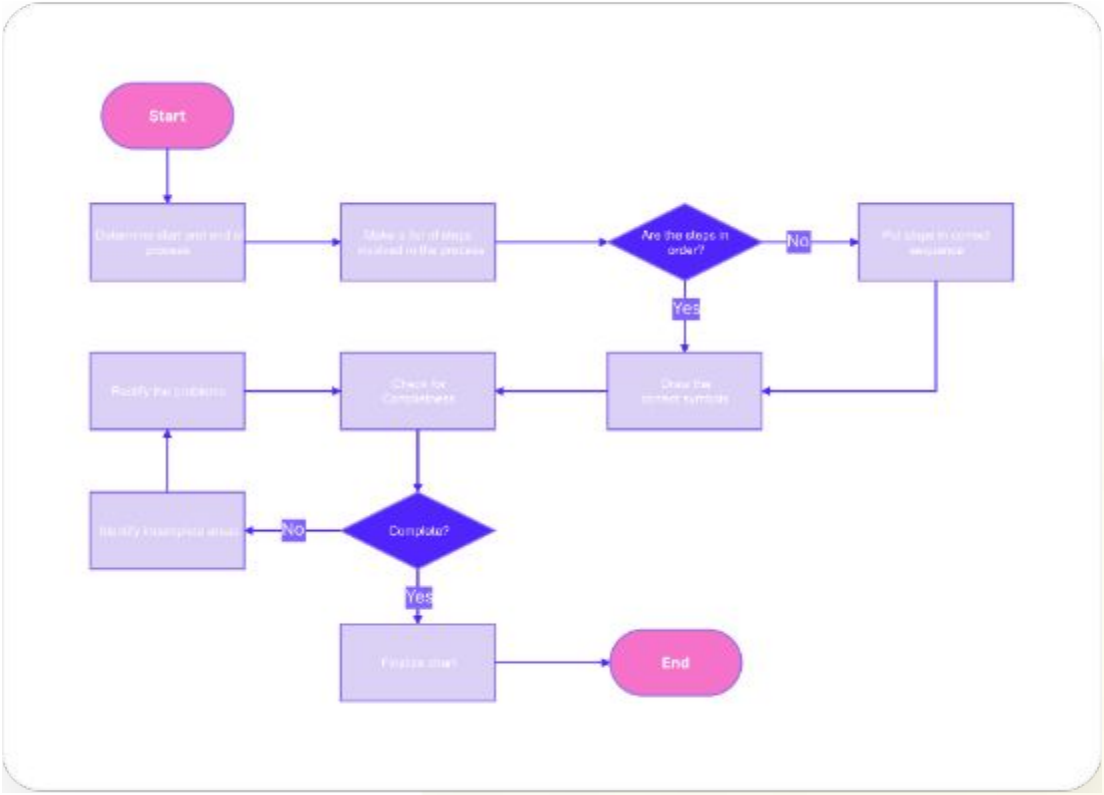
10 % 3 = 1 (Because 3 goes into 10 three times with 1 left over).

# Control Flow: Making Decisions

So far, our code runs top-to-bottom.

**Control Flow** allows the program to make decisions:

- "If the button is pressed, turn on the LED."
- "If the temperature is too hot, turn on the fan."



# The 'if' Statement

Checks a condition. If **TRUE**, executes the code block.

```
if (condition) {
// Code to run if
} true
```

```
if (temp > 3 ) {
// It's 0
  digitalWrit (1 , HIG );
} e           3   H
```

# How to Compare?

Used inside the () of the if statement.

| | |
|---|---|
| == | Equal to |
| != | Not Equal to |
| > | Greater than |
| < | Less than |

**Warning:** Don't confuse = (Assignment) with == (Comparison)!

# Else and Else If

```
if (temp > 30) {
  // Turn on AC
}
else if (temp < 10) {
  // Turn on Heater
}
else {
  // Do nothing, temp is fine
}
```

Allows you to handle multiple scenarios in a single logic block.

# Logical Operators

Combine multiple conditions.

**&&**

**AND**

True only if **BOTH** are true.

**||**

**OR**

True if **EITHER** is true.

**!**

**NOT**

Inverts the value (True becomes False).

# Loops: The 'for' Loop

Used to repeat a block of code a specific number of times.

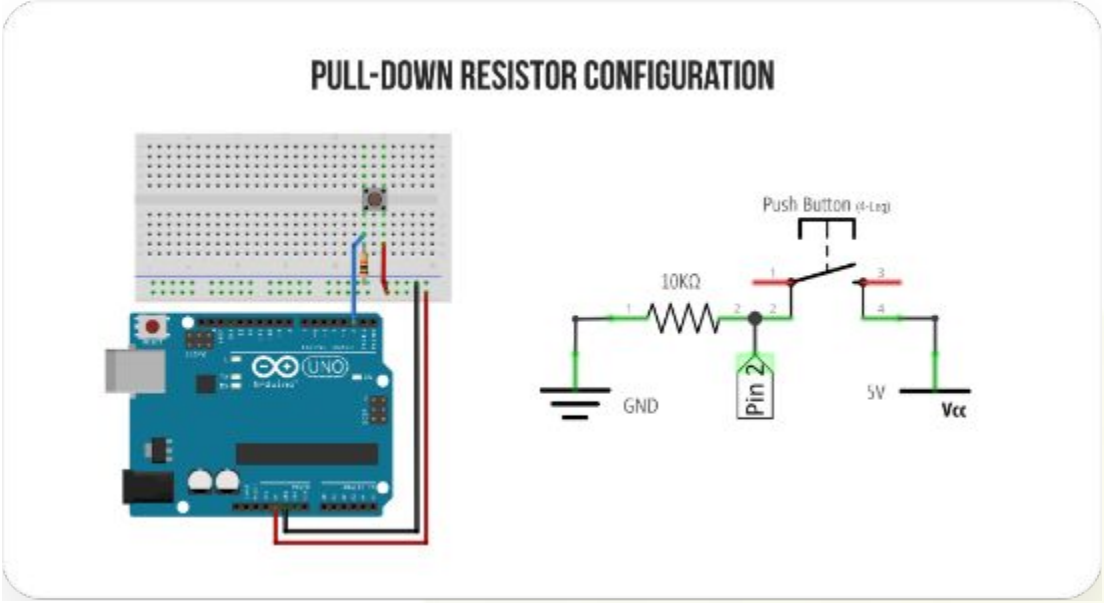
```
for (int i = 0; i < 5; i++) {
  // Run this 5 times
  digitalWrite(13, HIGH);
  delay(200);
  digitalWrite(13, LOW);
  delay(200);
}
```

## Structure:

1. **Initialize:** Start counter (int i = 0)
2. **Condition:** Stop when? (i < 5)
3. **Increment:** Count up (i++)

# Practical: Button Toggle Logic

```
if (digitalRead(button) == HIGH) {
  digitalWrite(led, HIGH);
} else {
  digitalWrite(led, LOW);
}
```




Connecting Logic to Hardware

# Summary



**Variables**

Named boxes to store data.

Must declare Type.



**Types**

int, float, bool define what data fits in the box.



**Flow**

if/else allows the code to make smart decisions.

# Questions?

"The only stupid question is the one you don't ask."

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# Image Sources



https://miro.medium.com/v2/resize:fit:1400/1*Px7h03Ih7B5QZu4KQpSEoQ.png

Source: studeappsblog.medium.com

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https://d8it4huxumps7.cloudfront.net/uploads/images/6470889b81f66_data_types_in_c_3.jpg

Source: unstop.com

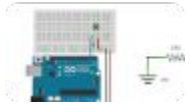
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https://images.edraw.ai/blogs/general/banner-flowchart.png

Source: www.edraw.ai

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https://makeabilitylab.github.io/physcomp/arduino/assets/images/ArduinoUno_Button_PullDownResistor_WiringDiagram.png

Source: makeabilitylab.github.io